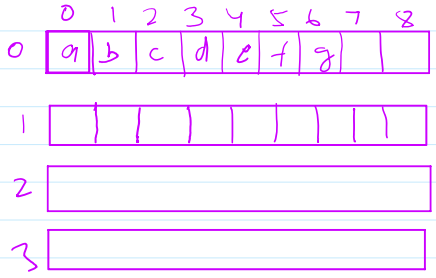
String Revision

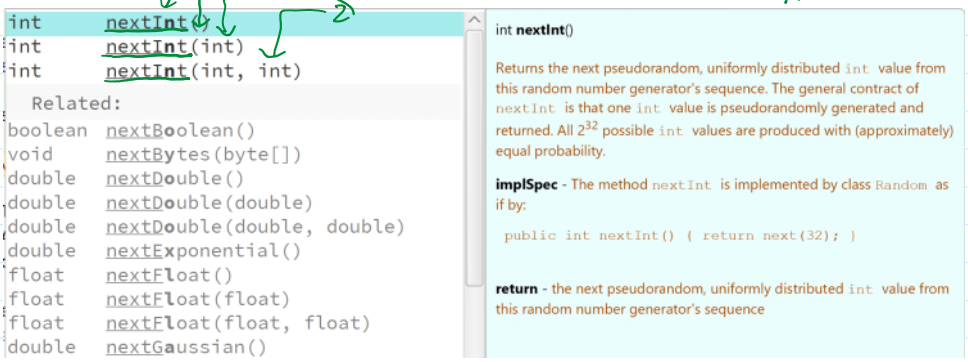
28 February 2026 09:55

In String object we store group of characters, and they are stored sequentially inside the String object.

```
String smallAlpha = "abcdefghijklmnopqrstuvwxyz";
String bigAlpha = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
String space = " ";
String newline = "\n";
String tabkey = "\t";
```

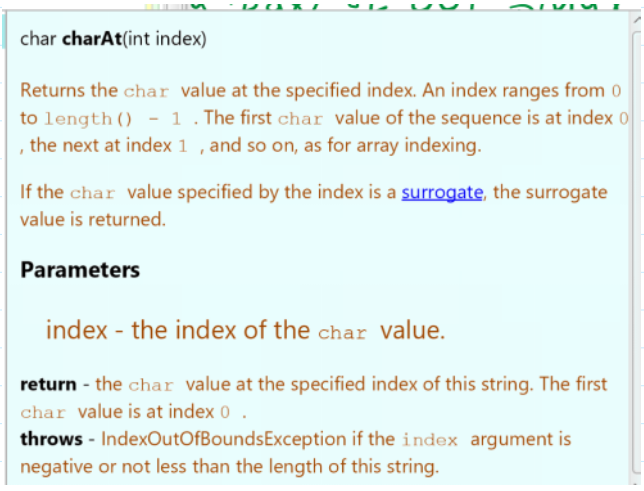


method name is same and number of formal parameters are different. This technique is called method overloading. It is a part of OOP pillar called Polymorphism.



https://docs.oracle.com/en/java/javase/25/docs/api/java.base/java/util/Random.html

Use this link to study more about nextInt() methods.




```
package StringClasses;
import java.util.*;
```

```java
/**
 * Write a description of class IntersetingStringCode here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class IntersetingStringCode
{
    public static void main(String[] args){
        Random r = new Random();
        String smallAlpha = "abcdefghijklmnopqrstuvwxyz";
        String bigAlpha = smallAlpha.toUpperCase();
        String space = " ";
        String tab = "\t";
        String enterKey = "\n";
        int i, sizeOfName, sizeOfSurname, numOfNames, j;
        String name, surName;
        Scanner sc = new Scanner(System.in);
        //Random name Name(4-10), Surname(4-10)
        System.out.print("\nHow many names you want to generate: ");
        numOfNames = sc.nextInt();
        for(i = 0; i < numOfNames; i++){
            sizeOfName = r.nextInt(4, 11);
            sizeOfSurname = r.nextInt(4, 11);
            //name
            name = ""+bigAlpha.charAt(r.nextInt(26));
            for(j = 1; j < sizeOfName; j++){
                name += smallAlpha.charAt(r.nextInt(26));
            }
            //surName
            surName = ""+bigAlpha.charAt(r.nextInt(26));
            for(j = 1; j < sizeOfSurname; j++){
                surName += smallAlpha.charAt(r.nextInt(26));
            }
            System.out.print(enterKey+ i+ tab + name + space + surName);
        }
    }
}


package StringClasses;
import java.util.Scanner;


/**
 * Write a description of class LongestWord here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class LongestWord
{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String sent, word="", longWord = "";
        int i, spcIndex;
        System.out.print("\fEnter a sentence: ");
```

" Tata football academy with play finals

i = 0 5 14 22 27 32 39

spcIndex = 4 13 21 26 31 38

word = ~~Tata~~ ~~football~~ ~~academy~~ with ~~play~~ finals

longWord = ~~Tata~~ football

```
        String sent, word="", longWord = "";
        int i, spcIndex;
        System.out.print("\fEnter a sentence: ");
        sent = sc.nextLine();
        sent = sent.trim() + " ";
        for(i = 0; i < sent.length(); i = spcIndex + 1){
            spcIndex = sent.indexOf(" ", i);
            word = sent.substring(i, spcIndex);
            if(word.length() > longWord.length()){
                longWord = word;
            }
        }
        System.out.print("\nThe longest word : " + longWord);
        System.out.print("\nThe length of the word is : " + longWord.length());
    }
}
```

Output:

```
Enter a sentence: tata football academy will play finals against mohan bagan

The longest word : football
The length of the word is : 8
```

```
package StringClasses;
import java.util.*;

/**
 * Write a description of class LongestWordSchoolTechnique here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class LongestWordSchoolTechnique
{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String sent, word="", longWord = "";
        int i, spcIndex;
        System.out.print("\fEnter a sentence: ");
        sent = sc.nextLine();
        sent = sent.trim() + " ";
        for(i = 0; i < sent.length(); i++){
            char c = sent.charAt(i);
            if(c != ' '){
                word += c;
            }
            else{
                if(word.length() > longWord.length()){
                    longWord = word;
                }
                word = ""; //empty the word
            }
        }
        System.out.print("\nThe longest word : " + longWord);
        System.out.print("\nThe length of the word is : " + longWord.length());
    }
}
```

Wrapper classes in Java
In Java, a wrapper class is a class that encapsulates (wraps) a primitive data type, into an object. This allows primitive values (int, care double, etc.) to be used where objects are required.

Primitive type	Wrapper Class
byte	Byte
short	Short
int	Integer
long	Long
float	Float
double	Double
char	Character
boolean	Boolean

Key features of Wrapper classes
1. **Object Representation-** Allows primitive to be treated as objects.
2. **Autoboxing:** Automatic conversion from primitive to wrapper object.
3. **Unboxing:** Automatic conversion from wrapper object to primitive.
4. **Utility methods:** Parsing, converting and comparing values.

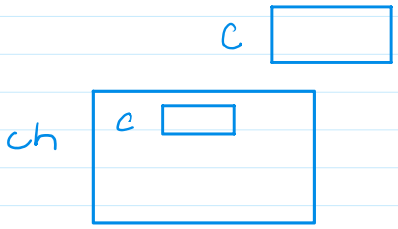
```
package StringClasses;
import java.util.*;


/**
 * Write a description of class Q12 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q12
{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String sent;
        System.out.print("\fEnter any sentence: ");
        sent = sc.nextLine().trim();
        int i;
        char c, nc='\0';
        sent = " " + sent;
        for(i = 0; i < sent.length() - 1; i++){
            c = sent.charAt(i);
            nc = sent.charAt(i+1);
            if(Character.isSpace(c) && Character.isLetter(nc)){
                System.out.print(Character.toUpperCase(nc));
            }
            else{
                System.out.print(Character.toLowerCase(nc));
            }
        }
    }
}
```

name of wrapperclass. functionname (parameters)

Note: Most of the function present inside wrapper class is static, hence it needs class reference not the object reference.

```
char c;
Character ch = c; //boxing
c = ch; //Unboxing
```




```
package StringClasses;
import java.util.*;


/**
 * Write a description of class Q12Alternative1 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q12Alternative1
{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String sent, word="";
        System.out.print("\fEnter any sentence: ");
        sent = sc.nextLine().trim();
        int i;
        char c, nc='\0';
        sent = sent+ " ";
        for(i = 0; i < sent.length(); i++){
            c = sent.charAt(i);
            if(!Character.isSpaceChar(c) ){
                word += c;
            }
            else{
                nc = Character.toUpperCase(word.charAt(0));
                System.out.print(nc+word.substring(1).toLowerCase()+c);
                word= "";
            }
        }
    }
}


package StringClasses;
import java.util.*;


/**
 * Write a description of class Q12Alternative2 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q12Alternative2
{
```

```
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String sent;
        System.out.print("\fEnter any sentence: ");
        sent = sc.nextLine().trim();
        int i;
        char c, nc='\0';
        sent = " " + sent;
        for(i = 0; i < sent.length() - 1; i++){
            c = sent.charAt(i);
            nc = sent.charAt(i+1);
            if(c == ' ' && nc >= 'a' && nc <= 'z'){
                System.out.print((char)(nc-' '));
            }
            else if (c != ' ' && nc >='A' && nc <= 'Z'){
                System.out.print((char)(nc + ' '));
            }
            else{
                System.out.print(nc);
            }
        }
    }
}
```

```
package StringClasses;
import java.util.*;


/**
 * Write a description of class Q13 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q13
{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String word;
        int i, asc;
        System.out.print("\fEnter any word: ");
        word = sc.next();
        for(i = 0; i < word.length(); i++){
            char c = word.charAt(i);
            asc = c;
            System.out.print("\nASCII of " + c + " = " + asc);
        }
    }
}
```

```
package StringClasses;
import java.util.*;


/**
 * Write a description of class Q14 here.
```

```
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q14
{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String sent, vowel = "aeiouAEIOU";
        System.out.print("\fEnter any sentence: ");
        sent = sc.nextLine();
        char c;
        int i;
        for(i = 0; i < sent.length(); i++){
            c = sent.charAt(i);
            if(vowel.indexOf(c) > -1){
                System.out.print("*");
            }
            else{
                System.out.print(c);
            }
        }
    }
}
```